AI Workflows for Empirical Research

General Setup: Claude Code & Codex

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The instructions file

- Agentic tools read a Markdown file at every session start — persistent project memory.
- Codex (+ ∼20 tools) call it `AGENTS.md`; Claude Code calls it `CLAUDE.md` (and also reads `AGENTS.md`).
- Open standard since Aug 2025 (Linux Foundation); 60,000+ repos. Just Markdown, no required fields.
- Like a "README for the agent": what you'd tell a new RA, written once.

▸ agents.md ▸ Claude Code memory

Using it well

- Keep it short (Anthropic: $<\sim$200 lines); put run/test commands at the top.
- Sections: overview, setup commands, conventions (e.g. cluster SEs), data-security notes.
- **MCPs** = connections to external tools/data; **skills** = reusable, on-demand instructions.
- Both run in a sandbox and can execute code + edit files — power and risk for reproducibility.

▸ AGENTS.md guide ▸ CLAUDE.md best practices ▸ Codex CLI guide

Skills: what they are

- A folder with a `SKILL.md` (YAML `name` + `description`, then Markdown instructions); can bundle scripts + resources.
- Like an onboarding guide for a repeatable task (e.g. "format regression tables").
- **Progressive disclosure**: metadata always loaded (~100 tokens); instructions load on trigger; bundled files/scripts load only when used.
- The `description` decides when Claude fires the skill — say *what* it does and *when* to use it.

▸ Agent Skills docs

Where to find & install them

- First-party (trusted): Anthropic's repo + pre-built doc skills (pptx, xlsx, docx, pdf).
- Community directories (vet first): awesome-claude-skills lists, marketplace sites.
- Install by surface: Claude Code = drop in `~/.claude/skills/` or `.claude/skills/`; claude.ai = upload zip; API = `/v1/skills`.
- Don't sync across surfaces; API sandbox has no network access.

anthropics/skills awesome-claude-skills

Are they safe?

- Anthropic: use skills only from trusted sources — treat like installing software.
- Research: prompt injection in $\sim$36% of skills studied; $\sim$91% of malicious ones bundle malware too.
- Audit every file (incl. scripts + secondary refs); flag external URL fetches; scope Claude's permissions.
- Not covered by zero-data-retention — relevant for confidential microdata.

▸ Snyk ToxicSkills ▸ Datadog Security Labs

MCPs in empirical research (I)

What they are, briefly: standardized connectors that let Claude reach your tools and data (databases, files, APIs, reference managers).

- **Data access:** pull series directly — e.g. FRED MCP exposes 800,000+ economic time series to Claude.
- **Literature:** search arXiv / OpenAlex / Semantic Scholar and push hits into Zotero for review.
- **Your project:** filesystem / database servers so Claude reads your data schema and files in place.

▸ FRED MCP ▸ Zotero MCP ▸ MCP Registry

MCPs in empirical research (II)

Workflow & caution:

- Chain steps: fetch data $\rightarrow$ clean $\rightarrow$ estimate $\rightarrow$ draft — without leaving the agent.
- Reproducibility: version control + issue trackers via MCP keep an audit trail of what changed.
- **Security:** a server is code with data + execution access; prefer local servers you control, scope credentials, vet third-party ones.
- For confidential microdata: avoid remote servers; treat community servers like any unvetted dependency.

▸ Tool poisoning (OWASP) ▸ AI for economists